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Full Stack Development / Further Web Programming Week 1

Topics Covered:

& Prerequisites knowledge
& Install VS Code, GitHub, and other software tools and prepare your work environment
& Simple web app using HTML, CSS, JavaScript

Activity 1: Investigate a popular web application

With your peers, discuss how web apps might be built and investigate:

- What front-end technology might it use?
- What back-end technology might it use?
- What databases might it use?
- Where might it be hosted?
- Can you find a systems architecture diagram of the app?
- What is nice about this web app? What is not so nice about it?

Activity 2: Hardware & CLI requirements

Please make sure you bring your own device to each practical class with good access to the Internet. You will need to bring a recent Windows laptop or a MacBook at least 16GB of RAM, an Intel Core i5/i7 (12th gen+) or an Apple M1/M2/M3 chip, and a fast SSD (512GB+). Older machines (3 years or older) may not cope with the heavy load of developing and serving React applications.

We will be using Command Line Interpreter to install software dependencies, run web servers, and some file / directory management. If you are using Windows, you need to be familiar with

basic CMD or PowerShell operations. If you are using Mac, you need to be familiar with MacOs (bash) commands.

Please see the following command line reference: SS64 Command line reference

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Activity 3: Revise HTML, CSS & JavaScript (Self-Exercise)

Review basic HTML, CSS, and JavaScript tutorials. References can be found in Module - Week 1 learning material:

& HTML: HTML Tutorial
& CSS: CSS Tutorial
& JavaScript: JavaScript Tutorial

It is crucial to be comfortable with these topics, as basic concepts will not be covered in this elective.

Activity 4: Install VS Code

Download Visual Studio Code (preferred IDE for this course) from VS Code Download. *Note:*

You may use other IDEs, but support will only be available for VS Code.

Activity 5: A bit about GitHub

As a developer, you will need to administer software repos on GitHub while collaborating with your peers or supervisors. Please read the following starting point: https://github.com/rmit-fsd-2026-s1/rmit-fsd-2026-s1-classroom-git-and-github-fundamentals-github-starter-course.git

You will start a portfolio of working programs you learn in the practical classes in order to do well in Assessment Task 3: Individual Programming Test that you will take in Week 12 practical class. HINT: The assessment will be mostly based on the prac materials and particularly the Weekly Practical Exercise.

Please create a GitHub account using your RMIT email ID and accept the invite to this Classroom repo to start practicing using with GitHub: https://classroom.github.com/a/dOxEsbot

Once accepted, you will be provided with your own dedicated repo to practice with GitHub commands, e.g. git pull, git push, etc. Alternatively, you can also use GitHub Desktop to get visual representation of your GitHub repositories and manage your repos more easily.

Activity 6 – **Weekly Practical Exercise**

Week 1: Create a simple web app using HTML, CSS, and JavaScript.

Target task completion: within 1 hour

You will keep collecting Weekly Practical Exercise from Week 1 until Week 11 in the same repository starting with the following exercise.

1. Accept the invite to the Weekly Practical Exercise repositor on GitHub Classroom: https://classroom.github.com/a/zjK-vxOW
2. In your laptop, using VS Code or File Explorer or CLI terminal, create a folder called Week 1. This folder must be synchronized with the remote repository in GitHub Classroom.
3. Using VS Code Start populating the folder with the files required to build the website. Starting with a README.md file will help you build good habits as a professional web developer.
4. You mission should you accept it 😊:

Using HTML, CSS, JavaScript, please design and build the interface similar to that of a popular web app you discussed earlier, while focusing on **registration form** with a specific feature that you consider having a good interactivity with users, e.g. web form with validations, field auto-complete, scroller, widgets, validation / error message feedback etc. At this stage you are not required to build the back-end side of things. Please focus on JavaScript events and event-handling mechanisms.

5. At the end of your work shift, don't forget to push your changes into the remote repo so your tutor can optionally review it and give you quick feedback in the next class if you need one.
6. After you get comfortable with these pre-requisite skills, feel free to try TypeScript and run the example codes from Lectorial 01.

Activity 7 – Professional Reflection

After today's prac, in your own time please summarise what you have learned so far. Consider keeping notes on popular applications like OneNote that you can keep with you even after you graduate from RMIT.

Good professionals should be able to reflect and:

- Explain the fundamental challenges and opportunities of working with existing web applications (CLO 1)
- Explain the fundamental challenges and opportunities of developing your own web applications (CLO 1)

You don't need to share your reflective notes with your peers / tutor but feel free to do it if you feel comfortable. Some key points to include in your reflection:

- Do you find lessons this week useful for your professional development at work?
- Will you be able to demonstrate some proficiency using the knowledge you learn this week?
- How will these knowledge skills be applicable in my current / future workplaces?
- What knowledge and skill gaps do I need to fill and prepare before the next class?

Postgraduate students, please apply deeper analysis / critical thinking.

Finally, check the Course Guide in the handbooks to make sure you are on track:

053172 - Full Stack Development (Undergraduate)

054228 - Further Web Programming (Postgraduate)

and regularly visit the course's Learning Management System (Canvas) for weekly updates: Full Stack Development (2610)